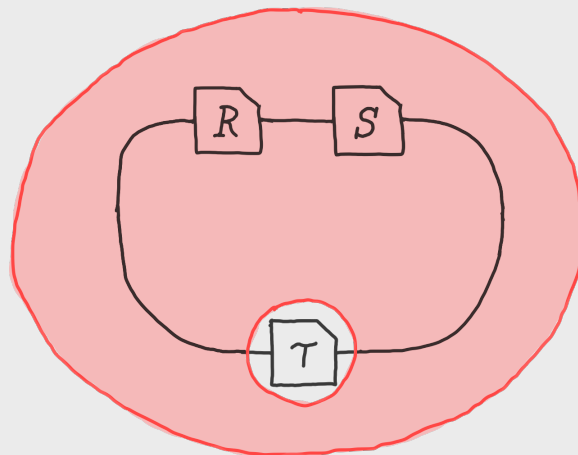
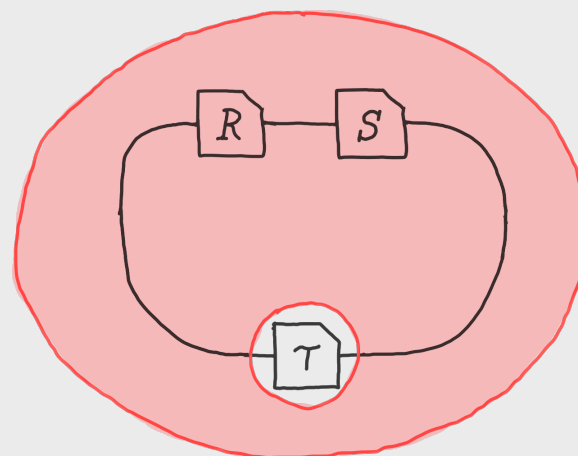
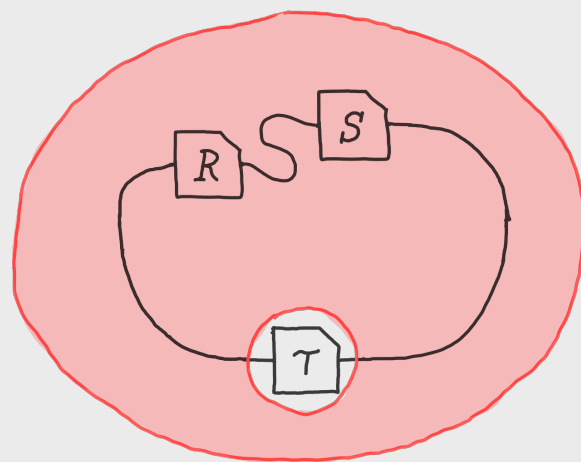
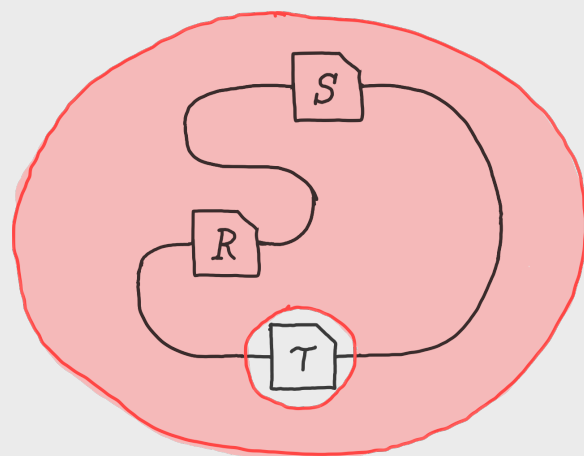


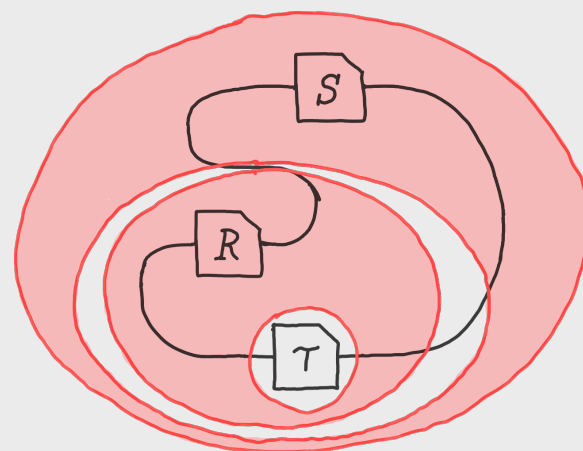


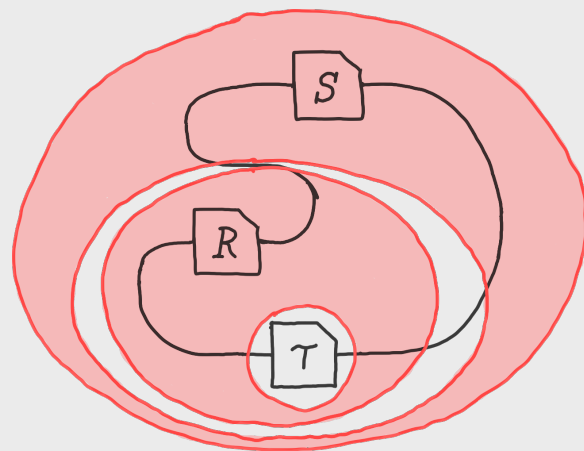
$$R; S \leq T$$



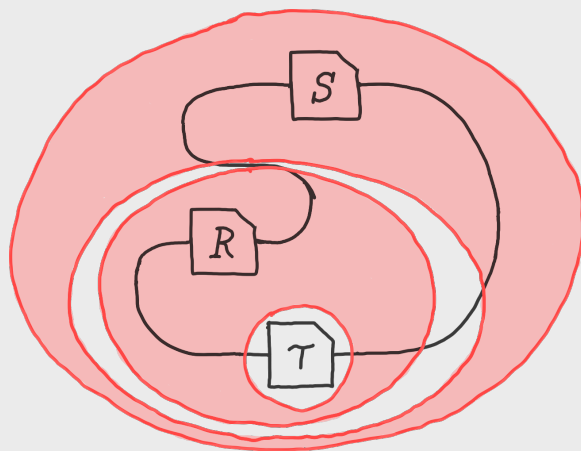








$$S \leq \overline{(\check{R}; \bar{T})}$$



$$S \leq \overline{(\check{R}; \bar{T})}$$

$$\parallel$$

$$R \setminus T$$

String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories

$$\text{---} \leq \text{---} \circ \text{---}$$

$$\text{---} \boxed{R} \text{---} \boxed{}$$

String Diagrams for Cartesian Bicategories

$$\text{---} \leq \text{---} \circ \text{---}$$

$$\text{---} \boxed{R} \text{---} \boxed{\text{---} \circ \text{---}}$$

String Diagrams for Cartesian Bicategories



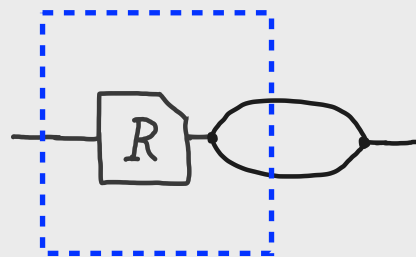
String Diagrams for Cartesian Bicategories



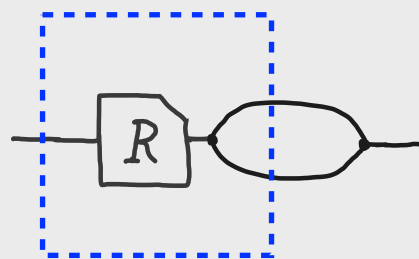
String Diagrams for Cartesian Bicategories



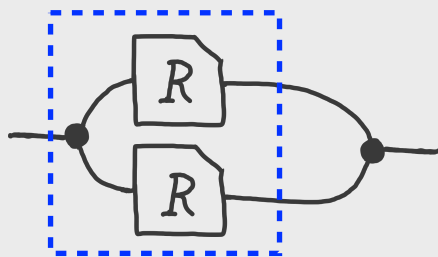
String Diagrams for Cartesian Bicategories



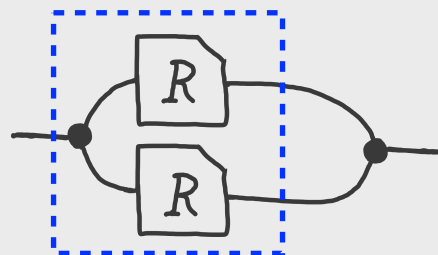
String Diagrams for Cartesian Bicategories



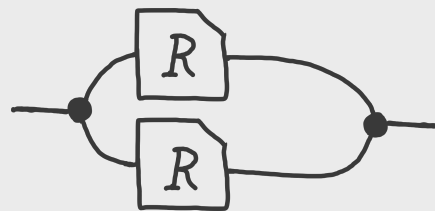
String Diagrams for Cartesian Bicategories



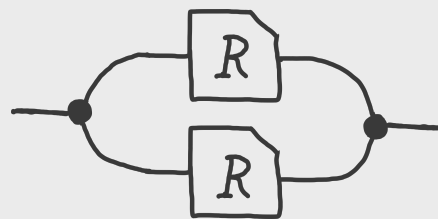
String Diagrams for Cartesian Bicategories



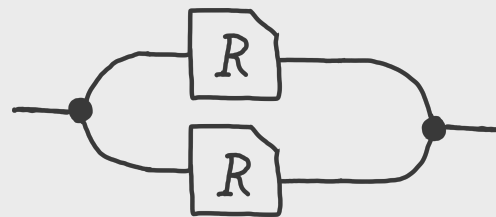
String Diagrams for Cartesian Bicategories



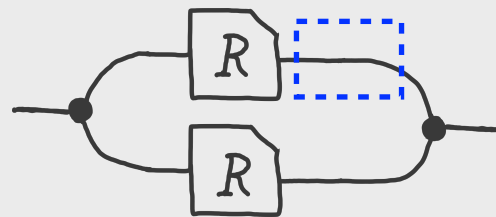
String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories

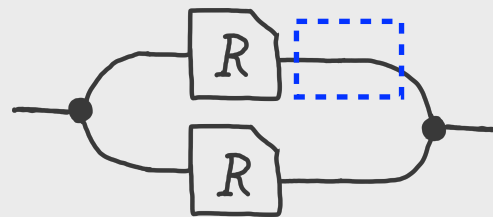


String Diagrams for Cartesian Bicategories



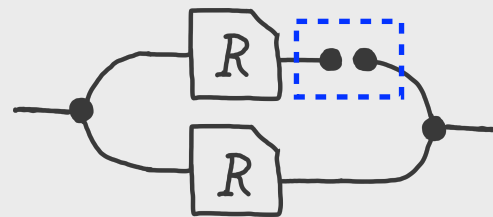
String Diagrams for Cartesian Bicategories

$$\text{---} \leq \text{---} \bullet \bullet \text{---}$$

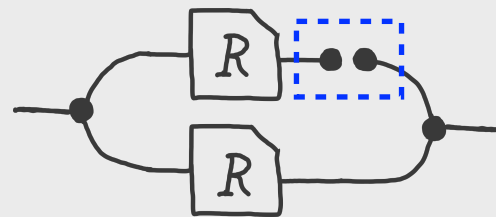


String Diagrams for Cartesian Bicategories

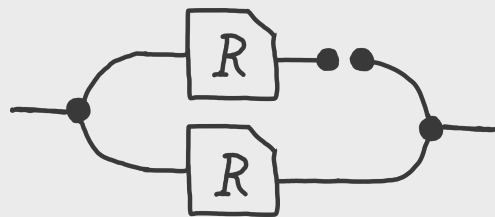
$$\text{---} \leq \text{---} \bullet \bullet \text{---}$$



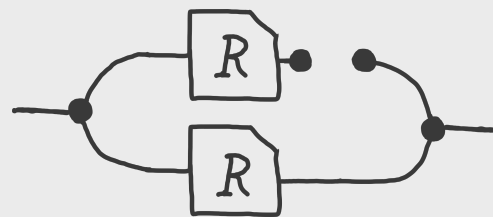
String Diagrams for Cartesian Bicategories



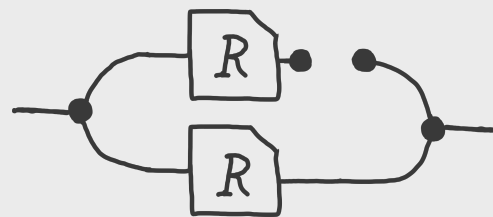
String Diagrams for Cartesian Bicategories



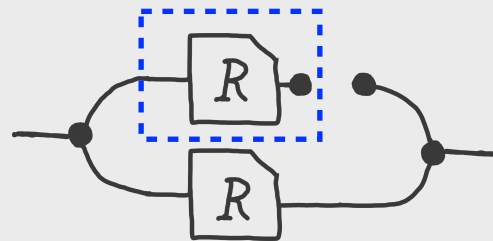
String Diagrams for Cartesian Bicategories



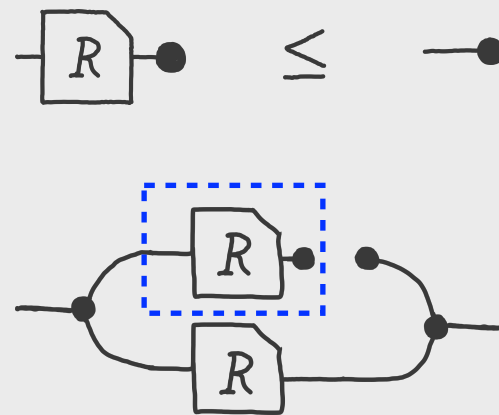
String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories

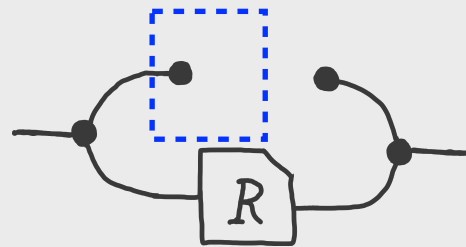


String Diagrams for Cartesian Bicategories

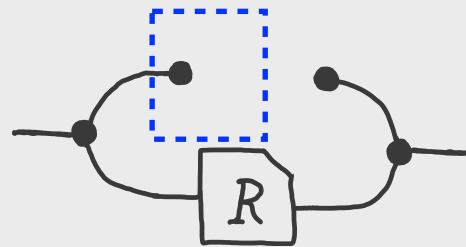


String Diagrams for Cartesian Bicategories

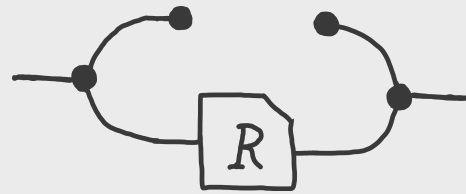
$$\text{---} \boxed{R} \bullet \leq \text{---} \bullet$$



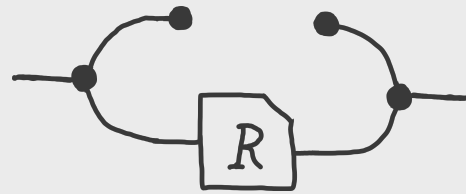
String Diagrams for Cartesian Bicategories



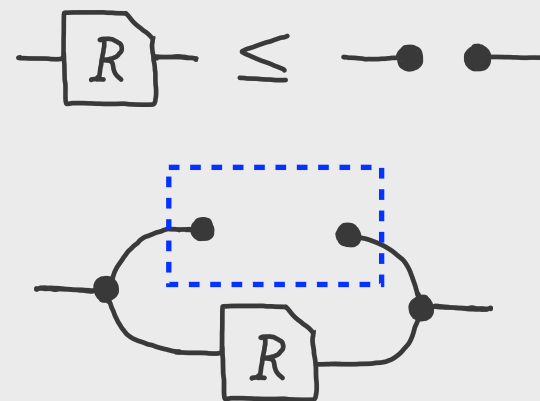
String Diagrams for Cartesian Bicategories



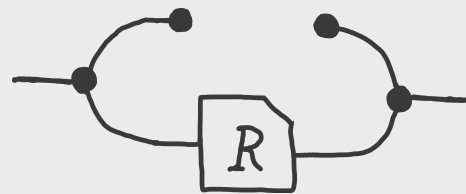
String Diagrams for Cartesian Bicategories



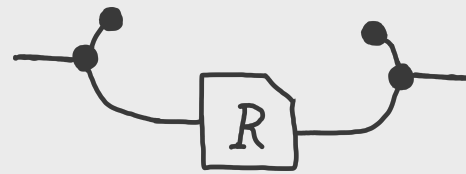
String Diagrams for Cartesian Bicategories



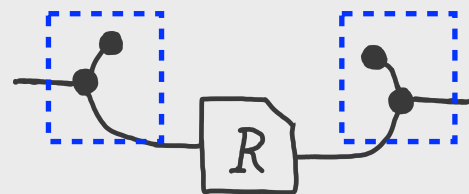
String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories

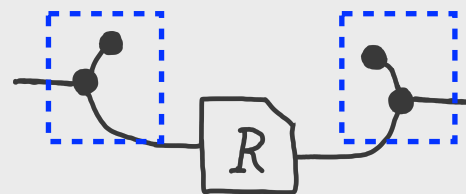


String Diagrams for Cartesian Bicategories



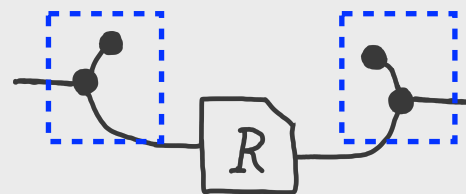
String Diagrams for Cartesian Bicategories


$$\text{multiplication node} = \text{horizontal line}$$



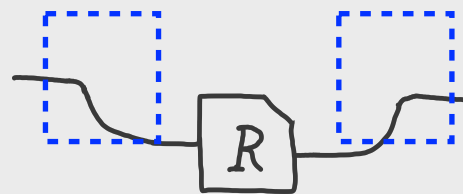
String Diagrams for Cartesian Bicategories


$$\text{multiplication node} = \text{horizontal line}$$

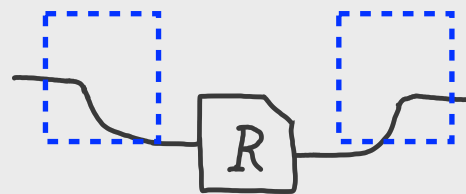


String Diagrams for Cartesian Bicategories

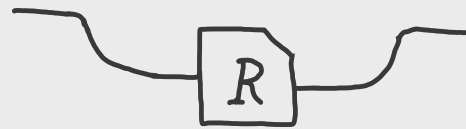

$$\text{multiplication} = \text{identity}$$



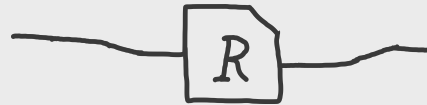
String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories



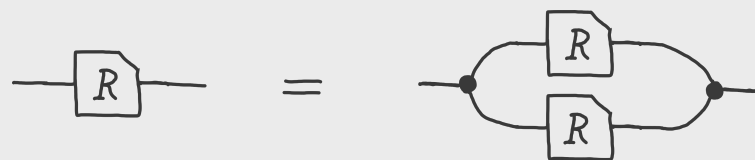
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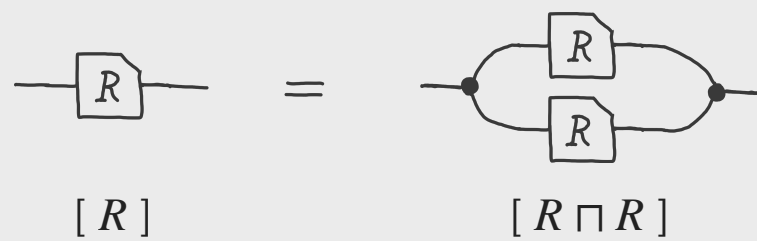
String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories



String Diagrams for Cartesian Bicategories



The diagram shows an equality between two string diagrams. On the left, a single horizontal line passes through a box labeled R . Below this diagram is the label $[R]$. On the right, two horizontal lines pass through two boxes labeled R stacked vertically. The lines are connected at both ends by curved arcs, forming a loop. Below this diagram is the label $[R \sqcap R]$. An equals sign $=$ is placed between the two diagrams.

$$[R] = [R \sqcap R]$$